

ANALGESIC LADDER FOR ACUTE PAIN MANAGEMENT

ANALGESIC LADDER FOR ACUTE PAIN MANAGEMENT				SEVERE		UNCONTROLLED
				Score 7-10		
				Regular	PRN	To refer to APS for: PCA or Epidural or other forms of analgesia
				Morphine	Additional Morphine	
				or IR Oxycodone	or IR Oxycodone	
				± PCM		
				± NSAIDs/ COX 2 inhibitor		

RECOMMENDED DRUG DOSAGES FOR ACUTE PAIN MANAGEMENT IN CHILDREN

(Review all regular use of analgesics no later than 72 hours after initiation)

Drug	Route	1 month-2 years	2-12 years	12- 18 years	> 18 years
		Dose and frequency			
Paracetamol	Oral	0 - 3 months: 15mg/kg 6 – 8 H (Max: 60mg/kg/day; if preterm 28-32 GA, max 30mg/kg/day) > 3months -12 years: 15mg/kg 4-6 H (Max: 90mg/kg/day or 4 grams/day)		500 mg - 1 gram 4-6 H (If non- obese ≥ 50 kg: 1 gram 4-6 H) (Max: 4 grams / day)	500 mg – 1gram 4-6 H (Max: 4 grams/day)
	Per rectal	0- 3 months: LD 30 mg/kg; MD : 20mg/kg 8 H (Max: 60mg/kg/day) > 3 month – 12 years : LD 40mg/kg ; MD : 15-20mg/kg 6 H (Max: 75mg/kg/day)			
	IV	Preterm neonate over 32/52 (GA) : 7.5mg/kg 8 H Term neonate & until 10 kg: 7.5 mg/kg 6-8 H (Max: 30mg/kg/day) > 10kg or child up to 50 kg: 15mg/kg 4- 6H (Max: 60 mg/kg/day, not exceeding 2 grams if < 33 kg, or 3 grams for 33-50 kg.)		If non obese > 50 kg : 1 gram 4 – 6 H (Max: 4grams/day) **Obese Children: 15mg/kg Adjusted body weight (Max: 4grams/day)	
Ibuprofen (NSAIDS)	Oral	< 3 months: not recommended > 3 months: 5mg/kg 6-8 H (Max: 20mg/kg/day) 6 months – 12 years: 5-10 mg/kg 6-8 H (Max: 30-40mg/kg/day or 1.2 grams/24 hours)		200 mg – 400 mg 4 - 6H (Max: 2.4 grams/day)	
Diclofenac (NSAIDS)	Oral	< 6 months: not recommended > 6 months or >10 kg: 0.3-1mg/kg 8 H (Max: 3 mg/kg/day up to 150 mg/day, whichever is less for 2 days)		Oral 25 - 50mg 8 H (Max: 3 doses/day)	
	Per rectal	> 1 year: 1 mg/kg 8-12 H (Max: 3 mg/kg/day up to 150 mg/day, whichever is less)		50-100 mg (oral to be started 18 hours after initial 100mg suppository)	
Mefenamic acid (NSAIDS)	Oral			> 14 years: 250mg 8 H	250-500mg 8 H
Celebrex (Cox2 inhibitor)	Oral	< 2 years: not recommended		> 2 years: weigh risks & benefits 10-25 kg: 50mg 12 H > 25 kg: 100mg 12 H	Celecoxib: 200mg 12H / 400mg daily (Max for 1 week)
Tramadol	Oral / IV	> 1 year: 0.5-1mg/kg 4-6 H (with caution) NB: For tonsillectomy max 1mg/kg /dose 6-8H		> 12years: 1mg/kg 4 - 6H (Max: 100mg/dose or 400mg/day)	50mg -100mg 6-8 H (Max 400mg/day)
Oxycodone	Oral	> 1 month: Immediate Release (IR Oxynorm) 0.1-0.2 mg/kg (max5mg) PRN or 4-6H (by APS / Palliative team) NB: Immediate release (IR Oxynorm) for acute pain; Slow release (SR Oxycontin) for severe background pain			IR Oxynorm 5-10mg 6 H SR Oxycontin 10 -20 mg BD
Morphine (Opioid μ- agonist)	Oral	> 1 month-1 year: 0.1mg/kg 4-6 H (for moderate – severe pain) (with caution) > 1 year: 0.1-0.2 mg/kg 4-6 H (for moderate pain) 0.2-0.4 mg/kg 4-6 H (for severe pain)		> 12yrs: 0.1- 0.3 mg/kg 4 -6 H (max 10-15mg/dose, up to 6x/ 24 hours)	
	SC	0.1-0.2 mg/kg < 6 months: (up to 4x/ 24 hours) > 6 months: (up to 6x/ 24 hours)	0.2 mg/kg (up to 6x/ 24 hours)	> 12yrs: 5 -10mg (up to 6x/ 24 hours)	
	IV	Slow titration : Refer iv morphine titration protocol 1-12 months: Max : 0.1 mg/kg (up to 4x /24 hours) > 1 year: Max : 0.1 mg/kg (up to 6x/ 24 hours)		Slow titration : Refer iv morphine titration protocol > 12yrs: 2.5 -10mg (up to 6x/ 24 hours)	
Naloxone(pure opioid antagonist)	IV	0.01 mg/kg IV (Max: 0.4 mg) may repeat every 2 minutes.			0.1 – 0.4 mg IV/IM/SC IV dose may repeat every 1-2 minutes
Ketamine	Oral/ IV	Oral: 2-10mg/kg; IV: 0.2-0.5 mg/kg (restricted use only for trained personnel) (To be used in combination with midazolam for procedural analgesia if Ketamine dose > 0.5 mg/kg)			

IV : intravenous, SC : subcutaneous, H : hourly, Max : maximum, LD : Loading dose, MD: Maintenance dose, IBW : Ideal body weight.

**For Obese Children, recommended adjustments for drug dosing:

Opioid: Ideal Body weight (IBW); Paracetamol and NSAID: Adjusted Body Weight= IBW+ 0.4 x (Actual BW-IBW)